Beyond Searching a Village: Learning to Recommend Diverse and Successful Collaborative Teams

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Abstract

Team recommendation involves selecting skilled experts to form an *almost surely* successful collaborative team, or refining the team composition to maintain or excel at performance. To address the tedious and error-prone manual process, computational approaches have been proposed, especially for web-scale social networks and widespread online collaboration and diversity of interactions. In this tutorial, with a brief overview of pioneering subgraph optimization approaches and their shortfalls, we deliberately focus on the recent learning-based approaches, with a particular in-depth exploration of graph neural network-based methods. More importantly, we then discuss team *refinement*, which involves structural adjustments or expert replacements to enhance team performance in dynamic environments. Finally, we discuss training strategies, benchmarking datasets, and open-source libraries, along with future research directions and real-world applications. Further resources are available at `https://fani-lab.github.io/OpeNTF/tutorial/recsys26`.

CCS Concepts

• General and reference → Surveys and overviews; **• Information systems → Recommender systems**; **Social recommendation**; **• Human-centered computing → Social recommendation**; **• Computing methodologies → Neural networks**; **• Theory of computation → Graph algorithms analysis**.

Keywords

Team Recommendation; Social Information Retrieval; `OpeNTF`; `Adila`

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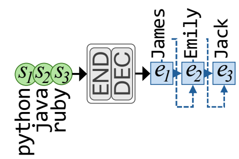

Figure 1: Team recommendation.

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[10 min] Pioneering Techniques
    [10 min] Subgraph Optimization Objectives and Techniques
[60 min] Learning Team Recommendation
[35 min] Team Refinement
    [15 min] Structural Adjustment
    [20 min] Membership Substitution
[20 min] Training Strategies
[10 min] Evaluation Methodologies
    [05 min] Datasets
    [05 min] Effectiveness/Efficiency
[15 min] Future Directions
    [05 min] Fair and Diverse Team Recommendation
    [05 min] Spatial Team Recommendation
    [05 min] Multi-Objective Neural Team Recommendation
[20 min] Real-World Applications
[10 min] OpeNTF and Adila
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Figure 2: Our tutorial's timeline.

1 Introduction

From Figure 1, team recommendation aims to automate forming collaborative teams of experts capable of successfully accomplishing a given task, and its applications span a wide range of domains. In emergency response, it involves assembling ad hoc teams of diverse professionals while accounting for expertise, communication efficiency, availability, and logistical constraints. In scientific publishing, it supports the assignment of peer reviewers, where identifying an appropriate review panel remains a bottleneck due to the manual selection process. In education, educators seek to partition students into collaborative teams that promote peer learning, engagement, and social skill development, a need that has become increasingly important with the growth of large-scale online classes. On online freelance platforms, where abundant information is available about experts' skills, past performance, and collaboration history, team recommendation enables forming effective teams for numerous projects with evolving requirements.

Historically, teams were formed manually based on human experience and instinct; a tedious, error-prone, and suboptimal process due to unfair biases, multiple criteria to optimize like budget, time and team size, and a large number of candidates, which makes it all the more difficult. Hence, researchers have long been developing computational models that learn relationships between experts and their skills to recommend *almost surely* successful teams, and/or refine existing teams to maintain or even improve their success.

2 Prior Tutorials

Despite the abundance, there had been *no* comprehensive tutorial on team recommendation methods until recently when we began to fill the gap by tutorials at `UMAP24`[1] [21], centered on *online*

[1] `https://www.um.org/umap2024/tutorials/`

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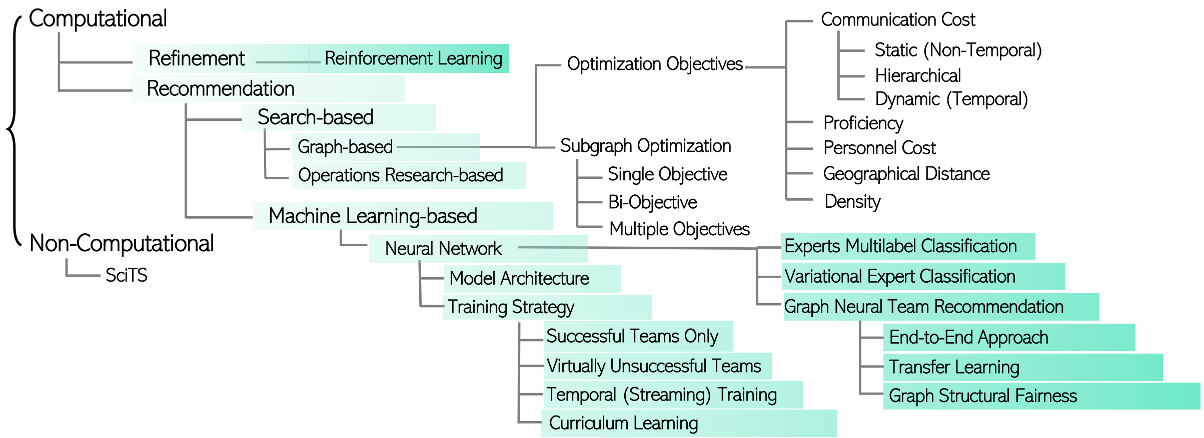

Figure 3: Taxonomy of team recommendation methods.

skilled users, SIGIR-AP24[2] [22], based on a novel taxonomy from a computational perspective with a focus on real-world applications, and at WSDM25[3] [23] and CIKM25[4] [19], to review seminal solutions with a special focus on the emerging graph neural network-based methods. In this tutorial, from Figure 2 and 3, we expand on them based on our recently published [20] and ongoing surveys of neural approaches that leverage graph neural networks [16, 27], sequence-to-sequence models and transformers [26]. Notably, we further investigate team refinement methods, such as expert replacement through reinforcement learning in dynamic sport teams or online video games [13, 28]. We also present methods to mitigate inherent biases via curriculum learning and loss regularization [4, 11, 15].

3 Targeted Audience

We target *beginner* or *intermediate* researchers, industrial practitioners with a broad interest in developing web-scale recommender systems, willing to have a whole picture of team recommendation techniques. Furthermore, we target the graph neural network community for a comprehensive review of subgraph optimization objectives and call them for further development of effective yet efficient graph machine learning frameworks with a special focus on team recommendation. Last, this tutorial enables organizations and practitioners to compare different approaches and readily pick the suitable one for their application to form or maintain successful teams. The target audience will be those *familiar* with graph theory and neural architectures. We provide sufficient details about advanced techniques so that the content becomes accessible to those with a fair knowledge of fundamentals.

4 Tutorial Summary (180 minutes)

Based on a taxonomy of computational methods, as shown in Figure 3, we start our tutorial with a brief introduction to the pioneering graph-based team recommendation algorithms [9], then continue to explore the learning-based team recommendation [1, 4, 7, 8, 10, 16–18, 24, 26] and team refinement methods [3, 13, 28, 29], focusing on modern graph neural networks and reinforcement learning. We explain strategies to train team recommenders, including negative sampling [6], temporal training [8], and curriculum learning [4]. We also discuss the benchmark datasets, what has been considered successful for a team as the ground truth, and metrics utilized to measure the quality of the recommended teams.

Toward the end, we discuss open issues and future directions, including:

- **Fair and Diverse Teams:** Existing methods heavily influenced by hidden *un*fair societal biases [11, 12, 15]. Meanwhile, social science research provides compelling evidence about the synergistic effects of diversity on team performance. In our tutorial, we introduce notions of fairness and protected attributes and study debiasing algorithms to mitigate the potential unfairness in the models' recommendations.
- **Spatial Team Recommendation:** In search of an optimal team, companies further look for experts in a region where the organization is *geographically* based. We bring forth the *spatial* team recommendation problem; that is, given a set of experts, skills and geolocations, the goal is to determine if the combination of skills and geolocations in forming teams has synergistic effects.
- **Multi-Objective Optimization:** In real-world scenarios, balancing multiple, often conflicting objectives (e.g., team effectiveness and experts' workload distribution) requires a training process guided by a loss function that explicitly accounts for multiple objectives. However, existing approaches commonly frame the problem as a classification or link prediction task, aim to maximize the coverage of the required skills, and overlook designing a task-aware loss function.

Moreover, we explain novel applications of team recommendation, including learning group recommendation [5, 14], peer-review assignments [2], and palliative care [25].

Finally, we present OpeNTF[5] [7, 10], a benchmark library for neural models that can efficiently preprocess large-scale datasets, can be easily extended or customized to new models, and is extensible to new datasets from other domains. We also introduce Adila[6] [11], which enables further post-processing reranking to the list of recommended experts to ensure a fair outcome, and viva*f*emme [15], an in-processing method for female-advocate loss regularization during model training.

[2]https://www.sigir-ap.org/sigir-ap-2024/tutorial/
[3]https://www.wsdm-conference.org/2025/tutorials/
[4]https://cikm2025.org/program/tutorials
[5]https://github.com/fani-lab/OpeNTF
[6]https://github.com/fani-lab/Adila

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